

Type-II Fatty-Acid Synthesis (FAS-II): A Dissociated, ACP-Dependent Enzyme Toolkit With Lineage-Swappable Parts

A commissioned review-style synthesis for a molecular-biology audience

1. Executive Summary

Type-II fatty-acid synthesis (FAS-II) is the bacterial (and plastid) route to *de novo* membrane acyl chains. Its defining architectural feature — the property that separates it cleanly from the animal/fungal type-I megasynthase (FAS-I) — is that every catalytic activity is carried out by a **discrete, individually encoded, freestanding enzyme** acting on substrates tethered to a small, soluble **acyl carrier protein (ACP)**. This "dissociated" organization is not a biochemical curiosity: it is the reason FAS-II is simultaneously (i) a validated, chemically rich antibacterial drug-target space, and (ii) a pathway whose individual steps can be, and repeatedly have been, replaced by non-homologous or divergent enzymes across bacterial lineages. The system therefore must be described as a *toolkit of interchangeable parts* rather than a single fixed machine.

The core logic is conserved and unambiguous. Acetyl-CoA carboxylase (the heteromeric AccABCD complex) commits acetyl-CoA to malonyl-CoA in the first regulated step; FabD transfers the malonyl group to holo-ACP; a chain is initiated; and a four-reaction elongation cycle — decarboxylative Claisen condensation (FabB/FabF), NADPH-dependent β -keto reduction (FabG), dehydration (FabZ/FabA), and enoyl reduction (FabI/FabK/FabL/FabV) — extends the chain two carbons per turn. A dedicated FabA/FabB branch introduces *cis* unsaturation anaerobically by isomerizing a *trans*-2 to a *cis*-3 decenoyl-ACP intermediate that FabB then elongates. This much is the shared skeleton.

Where the system varies, it varies **predictably and informatively**: at chain initiation (canonical FabH condensation vs. a genuinely FabH-independent Mad/YiiD malonyl-ACP decarboxylase route), at the dehydration/isomerization step (FabZ-only vs. dual-function FabA/FabN/FabQ, and the mycobacterial HadABC replacement), and — most dramatically — at enoyl

reduction, where four largely non-homologous isoforms with different cofactor preferences and inhibitor sensitivities occupy the same mechanistic slot. The overarching, cautionary conclusion of this review is that mechanistic claims validated in *Escherichia coli* should **not** be extrapolated uncritically to all bacteria: the pathway's modular design is precisely what permits deep lineage-specific substitution.

2. Definition and Biological Boundaries

What is inside the system

FAS-II, as scoped here, comprises the reactions that convert acetyl-CoA and malonyl-ACP into saturated and *cis*-monounsaturated acyl-ACP products destined for membrane phospholipid (and, in Gram-negatives, lipopolysaccharide) synthesis. The boundary set includes:

- **Malonyl-CoA formation** by acetyl-CoA carboxylase (AccABCD).
- **ACP activation and acyl carriage** — holo-ACP generated by phosphopantetheinyl transferase.
- **Malonyl-ACP loading** by FabD (malonyl-CoA:ACP transacylase).
- **Chain initiation** — FabH-dependent condensation and the FabH-independent Mad/YiiD decarboxylase route.
- **The four-reaction elongation cycle** — condensation (FabB/FabF), keto-reduction (FabG), dehydration (FabZ/FabA), enoyl reduction (FabI/FabK/FabL/FabV).
- **The FabA/FabB anaerobic *cis*-unsaturation branch.**

A hallmark of the entire system is that all intermediates remain **covalently tethered to ACP** via a phosphopantetheine thioester throughout elongation; the acyl chain is never released as a free acid between cycles.

What is often confused with FAS-II but should be treated separately

- **Type-I FAS (FAS-I).** The animal/fungal single-polypeptide (or fungal $\alpha 6\beta 6$) megasynthase performs the same chemistry but as covalently fused domains. FAS-II must not be "collapsed" into FAS-I; the essential distinction of this review is that FAS-II activities are physically separate proteins. This is what makes explicit enzyme-family and cofactor variants meaningful ([PMID: 35650664](#); [PMID: 23894064](#)).

- **Exogenous/environmental fatty-acid incorporation.** Bacteria can activate and incorporate host- or medium-derived fatty acids (via acyl-CoA, acyl-ACP, or acyl-phosphate intermediates depending on lineage). This *bypass* biology intersects FAS-II — and can defeat FAS-II-targeted drugs in some organisms — but is mechanistically distinct from *de novo* synthesis (PMID: 25648887; PMID: 28668270).
- **Phospholipid assembly downstream of FAS-II.** Enzymes such as PlsB/PlsC/PlsX/PlsY that transfer acyl chains onto the glycerophosphate backbone consume FAS-II products but are a separate pathway (PMID: 42324731).
- **β -oxidation / fatty-acid degradation.** The catabolic FadR-regulated route runs in the opposite direction on CoA-linked substrates and should not be merged with anabolic FAS-II (PMID: 17919287).
- **Mycolic-acid and specialized long-chain biosynthesis.** Mycobacterial FAS-II elongates very-long-chain products for mycolic acids and uses distinct dehydratases (HadAB/HadBC) and long-chain acyl-CoA carboxylases; these are lineage-specific elaborations of the core system, not the core itself (PMID: 17804795; PMID: 41843674).

Competing definitions

The main definitional tension in the literature is whether to treat FAS-II as a fixed enzyme list or as a functional logic realized by variable parts. This review takes the latter, more defensible position: the *slots* (initiate, condense, reduce β -keto, dehydrate, reduce enoyl) are conserved; the *occupants* are lineage-swappable. A secondary tension concerns the plastid FAS: plant and algal plastids run a bacterial-type dissociated FAS-II, so "FAS-II" is not strictly synonymous with "bacterial," and organelle context matters.

3. Mechanistic Overview

The committed entry and priming steps (obligatory)

1. **Malonyl-CoA formation (committed, regulated).** Acetyl-CoA carboxylase carboxylates acetyl-CoA in a biotin-dependent, two-half-reaction mechanism: biotin carboxylase (BC) carboxylates the biotin cofactor on the biotin-carboxyl-carrier protein (BCCP), and carboxyltransferase (CT) then transfers CO₂ to acetyl-CoA. In most bacteria this is a **heteromeric AccABCD** assembly (PMID: 39572150).

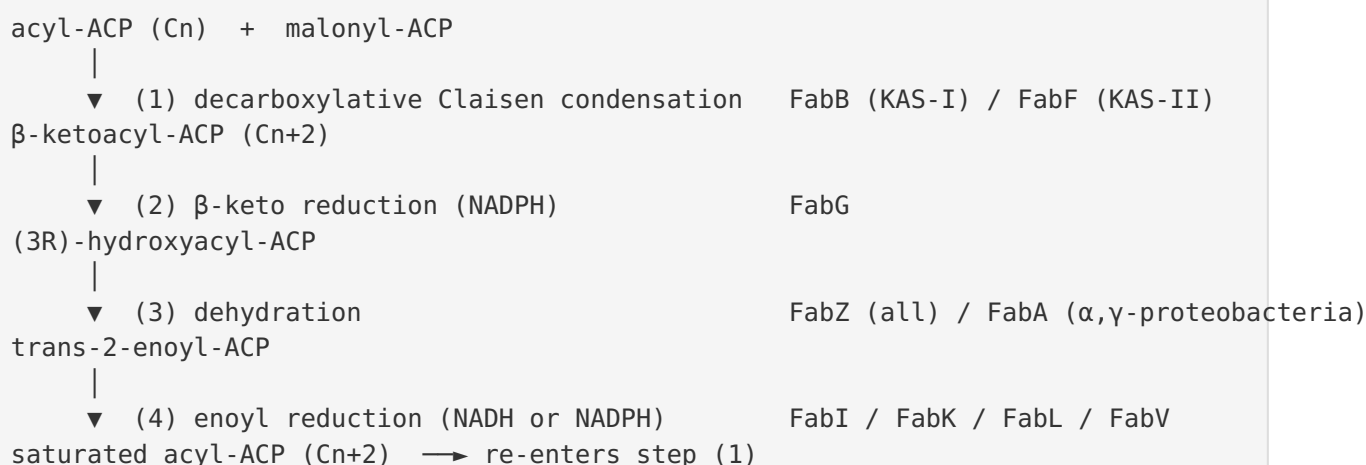
2. **Malonyl-ACP loading (obligatory).** FabD (malonyl-CoA:ACP transacylase) transfers malonate from CoA to holo-ACP, producing malonyl-ACP, the universal two-carbon donor for both initiation and elongation.

Chain initiation (obligatory step; alternative molecular routes)

- **Canonical route:** FabH (β -ketoacyl-ACP synthase III, KAS-III) condenses acetyl-CoA with malonyl-ACP to yield acetoacetyl-ACP (C4 β -ketoacyl-ACP).
- **Alternative route:** In organisms/strains lacking or bypassing FabH, the Mad/YiiD malonyl-ACP decarboxylase generates acetyl-ACP directly from malonyl-ACP; a FabB/FabF-type condensing enzyme then extends it. *E. coli* Δ fabH is viable specifically because of this route (PMID: 34801557).

The four-reaction elongation cycle (obligatory, iterative)

Each turn adds two carbons:



The FabA/FabB *cis*-unsaturation branch (conditional/accessory)

At the C10 stage, FabA can perform not only dehydration but also **isomerization** of *trans*-2-decenoyl-ACP to *cis*-3-decenoyl-ACP. FabZ cannot do this. FabB then elongates the *cis*-3 intermediate, committing the chain to the unsaturated arm and protecting the double bond from reduction. This is the classical anaerobic route to unsaturated fatty acids (UFAs) and is conditional on the FabA/FabB pairing (or functional equivalents such as FabM, FabN, or FabQ in other lineages) (PMID: 15371447; PMID: 23972938).

Which steps are obligatory vs. conditional vs. accessory

Step	Status	Notes
Malonyl-CoA formation (ACC)	Obligatory	First committed, regulated step
ACP activation	Obligatory	No carriage → no synthesis
FabD malonyl-ACP loading	Obligatory	Supplies the C2 donor
Chain initiation	Obligatory (route is conditional)	FabH or Mad/YiiD route
Condensation / keto-reduction / dehydration / enoyl-reduction	Obligatory (each cycle)	Enzyme identity is conditional
FabA/FabB isomerization branch	Conditional/accessory	Only when anaerobic UFA synthesis is used

4. Major Molecular Players and Active Assemblies

Finding F001 — FAS-II is a dissociated, ACP-dependent pathway distinct from FAS-I

Across authoritative reviews, bacterial FAS-II is defined by discrete, individually encoded enzymes acting on a small soluble ACP, in explicit contrast to the single multidomain megasynthase of animals and fungi. Because each of the four elongation activities is a separate protein, the pathway can host explicit enzyme-family and cofactor variants (FabB/FabF, FabZ/FabA, FabI/FabK/FabL/FabV). This dissociation is also why individual FAS-II enzymes are individually essential and individually druggable. As one review states, "*Bacterial type II fatty acid synthesis (FASII) is a vital aspect of bacterial physiology, not only for the formation of membranes but also to produce intermediates used in vitamin production*" (PMID: 35650664), and another emphasizes that "*Fatty acid synthesis type II (FASII) system enzymes are essential for bacterial membrane lipid biosynthesis and represent increasingly promising targets for the discovery of antibacterial agents*" (PMID: 23894064). The dissociated organization is thus both the defining structural fact and the source of the pathway's therapeutic relevance.

Finding F005 — Malonyl-CoA supply and ACP activation are the committed entry point and a shared cofactor hub

Bacterial acetyl-CoA carboxylase is heteromeric — biotin carboxylase, BCCP, and two carboxyltransferase subunits (AccABCD) — and operates through a biotin-dependent, two-half-reaction mechanism to make malonyl-CoA, the first committed and regulated step: *"In most Gram-positive and Gram-negative bacteria, the enzyme is composed of three proteins: biotin carboxylase, biotin carboxyl carrier protein and carboxyltransferase. The reaction consists of two half-reactions"* (PMID: 39572150). The isolated subunits are catalytically active, but the reconstituted complex shows allosteric behavior and is famously prone to dissociation: *"Historically the purification of active E. coli ACC complexes has been problematic due to the reported facile dissociation into subunits... Nevertheless, in reconstituted systems, the subunits appear to have allosteric interactions"* (PMID: 39572149). Downstream, FabD transfers malonate to holo-ACP. ACP itself is *"the universal factor required for fatty acid (FA) synthesis"* and is physically and functionally linked to other ancient essential processes such as iron-sulfur cluster biogenesis (PMID: 42294685), marking it as a deeply conserved cellular hub rather than a mere tether.

Finding F002 — Chain initiation has genuinely alternative molecular routes

Canonical initiation uses FabH (KAS-III) to condense acetyl-CoA with malonyl-ACP. However, *E. coli* Δ fabH strains remain viable because of a second route: the *yiiD* gene product (MadA) is a malonyl-ACP decarboxylase that generates acetyl-ACP for FabB/FabF condensation. The primary study is explicit: *"E. coli Δ fabH knockout strains are viable because of the yiiD gene that allows FabH-independent fatty acid synthesis initiation... we show the yiiD gene product is a malonyl-ACP decarboxylase (MadA)"* (PMID: 34801557). Moreover, FabH's own substrate range is tuned by a single active-site gatekeeper: in *E. coli* FabH a phenylalanine restricts the primer to acetyl-CoA, whereas in *M. tuberculosis* FabH the corresponding threonine admits longer acyl-CoA primers — *"In the ecFabH structure, this channel is blocked by a phenylalanine residue, which constrains specificity to acetyl-CoA, whereas in mtFabH, this residue is a threonine, which permits binding of longer acyl chains"* (PMID: 11278743). Initiation is therefore both mechanistically redundant (two routes) and tunable (gatekeeper residue), consistent with the outline's KAS-III substrate variants (FabH1 short-chain vs. FabH2 medium-chain).

Finding F003 — Unsaturation is introduced by the FabA/FabB branch; dehydratase family sets isomerase capability

In *E. coli*, FabA dehydrates (3R)-hydroxydecanoyl-ACP and isomerizes *trans*-2- to *cis*-3-decenoyl-ACP; FabB then elongates the *cis*-3 intermediate, committing it to the unsaturated arm. FabZ performs dehydration only. Structurally, FabA and FabZ share a near-identical His/Glu catalytic dyad, so **isomerase potential is dictated by the shape of the substrate-binding tunnel, not the catalytic residues**. This was demonstrated directly by domain swapping in *Enterococcus faecalis*: "*Substitution of the beta3 and beta4 strands of EfFabZ with the corresponding strands from EfFabN was necessary and sufficient to convert EfFabZ into an isomerase*" (PMID: 15980063). Other lineages reach the same outcome by different enzymes — for example, the dual-function FabQ: "*FabQ can not only replace the function of E. coli FabZ in vivo, but it also catalyzes the isomerization required for unsaturated fatty acid biosynthesis*" (PMID: 23972938). FabA's dual role and its more limited distribution relative to the universal FabZ are also well documented: "*FabA... has more limited distribution in nature and, in addition to dehydration, also carries out the isomerization of trans-2- to cis-3-decenoyl-ACP as an essential step in unsaturated fatty acid biosynthesis*" (PMID: 15371447).

Finding F004 — Enoyl reduction is the most variable step, with four non-homologous/divergent isoforms

The final reductive step is the single most variable slot in the pathway. It is carried out by at least four distinct enzyme families differing in fold, cofactor preference, and inhibitor sensitivity: **FabI** (short-chain dehydrogenase/reductase, SDR; triclosan-sensitive; includes *M. tuberculosis* InhA), **FabK** (a flavoenzyme, triclosan-resistant, streptococci), **FabL** (SDR, NADPH, *Bacillus*), and **FabV** (SDR-superfamily but only ~15% identity to FabI, ~402 aa, strong NADH preference, *Vibrio*). FabV was defined by complementing a conditionally lethal *E. coli* fabI mutant: "*FabV defines a new class of enoyl-ACP reductase... In contrast to FabI and FabL, FabV shows a very strong preference for NADH over NADP*" (PMID: 18032386). The four families are enumerated together in the literature: "*Enoyl-acyl carrier protein reductases (ENR), such as FabI, FabL, FabK, and FabV, catalyze the last reduction step in bacterial type II fatty acid biosynthesis*" (PMID: 29891603). Crucially, isoform identity has direct pharmacological consequences: "*some Gram-positive bacteria are naturally resistant to triclosan due to the presence of the triclosan-resistant enoyl-ACP reductase isoforms, FabK and FabL*" (PMID: 18032386). This step is also the most-exploited FAS-II drug target (triclosan → FabI; isoniazid → InhA).

Summary table of the major players

Slot	Canonical enzyme (E. coli)	Family / fold	Cofactor	Notable variants / lineage swaps
Malonyl-CoA formation	AccABCD	Biotin-dependent carboxylase (heteromeric)	ATP, biotin, HCO_3^-	Long-chain acyl-CoA carboxylase (Mycobacteria)
Malonyl-ACP loading	FabD	Acyltransferase	—	Broadly conserved
Initiation	FabH (KAS-III)	Thiolase-fold condensing enzyme	—	Mad/YiiD malonyl-ACP decarboxylase route; FabH gatekeeper tuning
Condensation	FabB (KAS-I) / FabF (KAS-II)	Thiolase-fold condensing enzyme	—	Clostridial FabF replaces both
β -keto reduction	FabG	SDR	NADPH	Broadly conserved (e.g., MabA in Mtb)
Dehydration	FabZ / FabA	Hot-dog fold hydratase	—	FabN, FabM, FabQ; HadAB/HadBC (Mtb)
Enoyl reduction	FabI	SDR	NADH/ NADPH	FabK (flavoenzyme), FabL (NADPH), FabV (NADH); InhA

5. Evolutionary and Cell-Biological Variation

Deep conservation vs. lineage-specific elaboration. The most ancient and conserved parts of the system are the ACP shuttle, the biotin-dependent carboxylation logic, the thiolase-fold condensing enzymes, and the SDR-fold β -keto reductase (FabG). ACP's physical linkage to iron-sulfur cluster biogenesis positions it as an especially ancient hub ([PMID: 42294685](#)). The *slots* — carboxylate, load, initiate, condense, reduce, dehydrate, reduce — are essentially universal across bacteria and plastids.

Where variation concentrates. Variation is not random; it accumulates at three points: 1. **Initiation** — FabH condensation vs. the Mad/YiiD decarboxylase route, plus FabH primer-length tuning (F002). 2. **Dehydration/isomerization** — FabZ-only vs. dual dehydratase/isomerases (FabA, FabN, FabQ), and the mycobacterial replacement of FabA/FabZ chemistry by HadAB/HadBC heterodimers of the hydratase-2 family ([PMID: 17804795](#)) (F003). 3. **Enoyl reduction** — four non-homologous/divergent families (FabI/FabK/FabL/FabV) with different cofactors and drug sensitivities (F004).

Unsaturation strategies differ by lineage. *E. coli* and other α/γ -proteobacteria use the FabA/FabB anaerobic branch. Streptococci use FabM (a *trans*-2 $\rightarrow$ *cis*-3 isomerase). *Aerococcus* uses the dual-function FabQ. *Clostridium acetobutylicum* makes the same UFA species yet lacks all known dedicated UFA genes, using a FabF homolog for both saturated and unsaturated synthesis while its lone FabZ cannot isomerize — evidence that at least one lineage introduces the double bond by a still-unidentified enzyme ([PMID: 19493359](#)). Aerobic organisms may instead install double bonds post-synthetically via O₂-dependent desaturases (a separate pathway, outside FAS-II proper).

Cell-biological/compartamental variation. In Gram-negatives, acyl-ACP products feed both phospholipid and LPS synthesis, whereas acyl-CoA (from exogenous fatty acids) cannot be used for LPS — creating a compartmentalized substrate-pool distinction ([PMID: 28668270](#)). In plant/algal plastids, the same dissociated FAS-II runs inside an organelle. Mycobacteria overlay a specialized long-chain FAS-II onto FAS-I output to build mycolic acids.

Regulatory variation. Transcriptional control is itself lineage-specific: *E. coli* uses FadR/FabR, *B. subtilis* uses FadR(YsiA)/FapR (with malonyl-CoA as the FapR effector), and *S. pneumoniae* uses FabT — different regulators governing the same metabolic logic ([PMID: 17919287](#)).

Best representatives of ancestral roles. When a family has expanded, the broadest-distribution, minimal-function member is the better proxy for the ancestral role: **FabZ** (pure dehydratase, universal) over FabA (dehydratase+isomerase, restricted) for dehydration; **FabG** for keto-reduction; and, among condensing enzymes, the KAS-I/KAS-II thiolase fold for the ancestral condensation chemistry, with FabH (KAS-III) representing a specialized initiation offshoot.

6. Constraints, Dependencies, and Failure Modes

Obligatory ordering. The chemistry enforces a strict order within each cycle: condensation must precede keto-reduction (no β -keto group exists until condensation), keto-reduction must precede dehydration (no 3-hydroxyl to eliminate otherwise), and dehydration must precede enoyl reduction (no *trans*-2 double bond to reduce otherwise). The saturated product must be regenerated before the next condensation. These dependencies are captured directly in the pathway's known relationships (condensation $\rightarrow$ keto-reduction $\rightarrow$ dehydration $\rightarrow$ enoyl-reduction $\rightarrow$ re-entry).

Timing constraint on unsaturation. The FabA-catalyzed *trans*-2 $\rightarrow$ *cis*-3 isomerization is a **kinetically gated, chain-length-specific branch point** at $\sim$ C10. If FabI (enoyl reductase) reduces the *trans*-2-decenoyl-ACP before FabA isomerizes it, the double bond is lost and the chain proceeds saturated. Commitment to unsaturation thus requires that FabB capture the *cis*-3 intermediate — a mutually exclusive fork between the saturated and unsaturated arms at that single stage.

Substrate tethering. All intermediates are ACP-bound; loss of ACP activation (phosphopantetheinylation) is catastrophic and upstream of everything. This dependency, plus ACP's link to Fe–S biogenesis, makes ACP a single point of failure ([PMID: 42294685](#)).

Cofactor and inhibitor constraints. The enoyl-reduction step's cofactor identity (NADH vs. NADPH) is enzyme-family-specific (F004), so redox-cofactor availability constrains which isoform can operate. Inhibitor sensitivity is likewise family-specific: triclosan inhibits FabI/InhA but not FabK/FabL, so an organism's isoform complement determines intrinsic resistance ([PMID: 18032386](#); [PMID: 29891603](#)).

Bypass as a systemic failure mode for therapeutics. In *Lactobacillales*, exogenous fatty acids fully suppress FAS-II, allowing membrane synthesis entirely from environmental lipids and defeating FAS-II inhibitors. In most other bacteria, FAS-II inhibition **cannot** be bypassed because essential intermediates (e.g., β -hydroxyacyl-ACP for LPS) cannot be obtained from the host ([PMID: 25648887](#); [PMID: 28668270](#)). This rules out the otherwise plausible assumption that all bacteria can escape FAS-II inhibition by scavenging.

Evidence that rules out otherwise-plausible paths. The *E. coli* Δ fabH viability result rules out the assumption that FabH is strictly indispensable for initiation ([PMID: 34801557](#)). The FabZ $\rightarrow$ FabN β -strand swap rules out the assumption that isomerase activity requires special catalytic residues ([PMID: 15980063](#)). The *Clostridium* result rules out the assumption that all bacteria use one of the known UFA enzymes ([PMID: 19493359](#)).

7. Controversies and Open Questions

1. **Is FAS-II a validated broad-spectrum drug target, or is bypass a fatal flaw?** The field disagreed sharply after reports that exogenous fatty acids rescue FAS-II inhibition. The current reconciliation is nuanced: bypass is real and clinically relevant in *Lactobacillales*, but most Gram-negative and many Gram-positive pathogens cannot bypass, so FAS-II remains targetable in a broad but not universal set ([PMID: 25648887](#); [PMID: 28668270](#)). Claims should specify the organism.
 2. **How is unsaturation introduced in lineages lacking known UFA enzymes?** *Clostridium acetobutylicum* synthesizes *E. coli*-type UFAs without any recognized UFA gene and with a FabZ that lacks isomerase activity, implying a "novel and unknown enzyme" ([PMID: 19493359](#)). This mechanism remains unidentified and is a significant open question.
 3. **Extrapolation risk across organisms.** Much mechanistic detail is anchored in *E. coli* (FabA/FabB, FabI). Extending these models to Gram-positives, Mycobacteria, or plastids risks conflating non-comparable systems, given the documented substitutions at initiation, dehydration, and enoyl reduction. This is an evidence-quality caution rather than a resolved controversy.
 4. **ACC assembly and regulation.** The bacterial ACC's facile dissociation and allosteric behavior complicate structural and kinetic interpretation; recent cryo-EM work is only now clarifying the reconstituted complex ([PMID: 39572149](#); [PMID: 39572150](#)). Whether the *in vivo* complex is a stable holoenzyme or a dynamic assembly is not fully settled.
 5. **The physiological weight of the Mad/YiiD route.** MadA rescues Δ fabH, but its native flux contribution when FabH is present, and its distribution across Proteobacteria, are incompletely mapped ([PMID: 34801557](#)).
 6. **Reliance on computational inhibitor studies.** A substantial fraction of recent FAS-II literature is docking/MD-based (InhA, FabI phytochemical screens). These provide hypotheses, not mechanistic proof, and should be weighted as indirect evidence pending biochemical validation ([PMID: 41921200](#); [PMID: 41780347](#)).
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8. Key References

PMID	Relevance to this review
35650664	Defines FAS-II as the essential bacterial membrane-lipid (and vitamin-intermediate) pathway; supports F001
23894064	FAS-II enzymes individually essential/targetable; supports dissociated organization (F001)
34801557	Identifies YiiD/MadA as malonyl-ACP decarboxylase enabling FabH-independent initiation (F002)
11278743	Structural basis (Phe vs. Thr gatekeeper) for FabH primer specificity (F002)
15980063	FabZ→FabN β -strand swap localizes isomerase determinant to substrate tunnel (F003)
23972938	FabQ as a dual dehydratase/isomerase alternative route (F003)
15371447	FabZ structure; FabA's dual role and limited distribution (F003)
18032386	FabV as a fourth NADH-preferring enoyl reductase class; triclosan resistance (F004)
29891603	Enumerates FabI/FabL/FabK/FabV families (F004)
39572150	Heteromeric AccABCD architecture and two-half-reaction mechanism (F005)
39572149	Dissociable, allosteric bacterial ACC subunits (F005)
42294685	ACP as universal carrier linked to Fe-S biogenesis (F005)
17804795	Mycobacterial HadAB/HadBC dehydratases replace FabA/FabZ chemistry
19493359	<i>Clostridium</i> makes UFAs via an unidentified enzyme; FabF replaces FabB/FabF
25648887	Host-lipid uptake and FAS-II bypass; therapeutic boundary
28668270	Lineage-specific exogenous fatty-acid activation; substrate-pool compartmentalization
17919287	Lineage-specific transcriptional regulation (FadR/FabR/FapR/FabT)
41843674	Mycobacterial long-chain acyl-CoA carboxylase structure
42324731	PlsB downstream phospholipid assembly (boundary case)

Limitations and Knowledge Gaps

This review is a literature synthesis and did not generate new experimental data. Several limitations should be made explicit:

- **Organism bias.** The mechanistic backbone is heavily *E. coli*-derived. Findings F002 and F003 rest substantially on *E. coli*/enterococcal work; their generality is inferred, not exhaustively verified across all bacterial phyla.
- **Incomplete variant coverage.** The outline's KAS-III sub-variants (FabH1 short-chain vs. FabH2 medium-chain) and the FabL "NADPH SDR" class are supported here largely by structural/gatekeeper reasoning and family enumeration rather than by side-by-side kinetic comparisons within a single organism.
- **Unresolved *Clostridium* mechanism.** A confirmed lineage introduces *cis* unsaturation by an enzyme that has not been identified — a genuine gap in the model.
- **ACC assembly uncertainty.** The physiological quaternary state of bacterial ACC remains partially unresolved.
- **Heavy reliance on computational inhibitor studies** in the most recent literature, which are hypothesis-generating rather than mechanistically definitive.

Proposed Follow-up Experiments / Actions

1. **Identify the *Clostridium acetobutylicum* UFA enzyme.** Combine comparative genomics of UFA-producing anaerobes lacking FabA/FabM/FabN/FabQ with heterologous complementation of an *E. coli* fabA(Ts) strain to isolate the missing isomerase activity.
2. **Quantify Mad/YiiD flux in wild-type backgrounds.** Use ¹³C-malonate isotope tracing in *E. coli* wild-type vs. ΔyiiD vs. ΔfabH to measure the native contribution of the decarboxylase initiation route.
3. **Systematic cross-lineage cofactor mapping of enoyl reductases.** Purify FabI, FabK, FabL, and FabV in parallel and measure NADH/NADPH kinetics and triclosan IC₅₀ under identical conditions to convert family-level generalizations into quantitative comparisons.
4. **Structural characterization of the reconstituted AccABCD holoenzyme** by cryo-EM under near-physiological conditions to resolve the assembly/allostery controversy.
5. **Test bypass susceptibility across a pathogen panel.** Empirically classify clinically important species as bypass-capable vs. bypass-incapable under FAS-II inhibition plus exogenous fatty acids, to define the true therapeutic window.

6. **Domain-swap generalization.** Extend the FabZ↔FabN β -strand swap logic across dehydratase families (FabA, FabQ, HadAB) to test whether substrate-tunnel geometry universally predicts isomerase capability.

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